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Air conditioner

Abstract

An air conditioner is provided, in which a control box is disposed at a boundary between an indoor main body configured to blow indoor air and an outdoor main body configured to blow outdoor air, and the outdoor air may be introduced into the control box. Accordingly, even if it rains, the air conditioner may prevent direct exposure of the control box and/or a heat sink of the control box to rainwater and facilitate heat dissipation of control components disposed in the control box configured to perform inverter control.

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Patent No.	Application Date	Country	CPC
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10-2019-0010240	12/2018	KR	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application claims priority to and the benefit of Korean Patent Application No. 10-2020-0020926, filed in Korea on Feb. 20, 2020, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

1. Field

(2) An air conditioner is disclosed herein.

2. Background

(3) Air conditioners may cool, heat, or purify air in a target space, or perform a humidification

operation or a dehumidification operation on the target space. The air conditioner may include a compressor, a condenser, an expansion device, and an evaporator to cool the air in the target space, and the compressor, the condenser, the expansion device, and the evaporator are sequentially connected to a pipe. Refrigerant may circulate in the compressor, the condenser, the expansion device, and the evaporator through the pipe.

(4) Air conditioners may include split-type air conditioners and integrated-type air conditioners. The split-type air conditioner may include an indoor unit located in an indoor space to discharge conditioned air into an indoor space and an outdoor unit connected to the indoor unit through a pipe and located in an outdoor space. The indoor unit may include an indoor heat exchanger and the outdoor unit may include an outdoor heat exchanger. When the split-type air conditioner performs a cooling operation, the outdoor heat exchanger may function as a condenser and the indoor heat exchanger may function as an evaporator. When the split-type air conditioner performs a heating operation, the indoor heat exchanger may function as the condenser and the outdoor heat exchanger may function as the evaporator.

(5) For the integrated-type air conditioner, the condenser and the evaporator are accommodated together in a case. The condenser is disposed in an outdoor portion of the case to exchange heat with outside air and the evaporator is disposed in an indoor portion of the case to exchange heat with indoor air.

(6) Korean Patent Publication No. 10-2019-0010240 (hereinafter, “related art document 1”), which is hereby incorporated by reference, discloses an integrated-type air conditioner. For the air conditioner according to related art document 1, a heat sink may be placed on a main printed circuit board (PCB) of a control box, may be exposed to an outside of the control box, and the control box may be disposed in an outdoor space to dissipate heat generated from components mounted on the main PCB of the control box, thereby preventing overheating of the components or the substrate. According to the related art document 1, the control box is located in the outdoor space for heat dissipation. Accordingly, if it rains, the heat sink may be directly exposed to rainwater, thereby degrading insulation performance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Embodiments will be described in detail with reference to the following drawings in which like reference numerals refer to like elements, and wherein:
- (2) FIG. 1 shows an example of an air conditioner located in a wall;
- (3) FIG. 2 is a schematic perspective view of an air conditioner according to an embodiment in a case;
- (4) FIG. 3 is a schematic perspective view of the air conditioner of FIG. 2, when viewed from the front;
- (5) FIG. 4 is a schematic perspective view of the air conditioner of FIG. 2, when viewed from the rear;
- (6) FIG. 5 is a schematic exploded perspective view of the air conditioner of FIG. 2;
- (7) FIG. 6 is a schematic perspective view of a control box of an air conditioner according to an embodiment;
- (8) FIG. 7 is a schematic exploded perspective view of the control box of FIG. 6;
- (9) FIG. 8 is a schematic side view of the control box of FIG. 6; and
- (10) FIGS. 9 to 11 shows air flow at an outdoor portion of an air conditioner.

DETAILED DESCRIPTION

- (11) Embodiments will be described with reference to the accompanying drawings, such that a person having ordinary knowledge in the art to which the embodiments pertain may easily

implement the technical idea. Description of well-known technology relating to embodiments may be omitted if it unnecessarily obscures the gist. One or more embodiments are described with reference to the accompanying drawings. The same or like reference numerals may be used to refer to the same or like components.

(12) It will be understood that, although the terms “first”, “second”, and the like may be used herein to describe various components, however, these components should not be limited by these terms. These terms are only used to distinguish one component from another component. Thus, a first component may be a second component unless otherwise stated.

(13) In this document, the terms “upper,” “lower,” “on,” “under,” or the like are used such that, where a first component is arranged at “an upper portion” or “a lower portion” of a second component, the first component may be arranged in contact with the upper surface or the lower surface of the second component, or another component may be disposed between the first component and the second component. Similarly, where a first component is arranged on or under a second component, the first component may be arranged directly on or under (in contact with) the second component, or one or more other components may be disposed between the first component and the second component.

(14) Further, the terms “connected,” “coupled,” or the like are used such that, where a first component is connected or coupled to a second component, the first component may be directly connected or able to be connected to the second component, or one or more additional components may be disposed between the first and second components, or the first and second components may be connected or coupled through one or more additional components.

(15) Unless otherwise stated, each component may be singular or plural throughout the disclosure.

(16) In some examples, singular expressions used include plural expressions unless the context clearly indicates otherwise. Terms such as “including” or “comprising” should not be construed as necessarily including all of the various components, or various steps described in the present disclosure, and terms such as “including” or “comprising” should be construed as not including some elements or some steps or further including additional elements or steps.

(17) Unless otherwise stated, “A and/or B” means A, B or A and B. Unless otherwise stated, “C to D” means “C or more and D or less”.

(18) Hereinafter, an air conditioner and a method for controlling the air conditioner according to embodiments are described.

(19) FIG. 1 shows an example of an air conditioner **100**. As shown in FIG. 1, the air conditioner **100** according to an embodiment may be an integrated-type air conditioner. In this structure, a condenser and an evaporator of the air conditioner **100** according to an embodiment may each be accommodated inside of a case. The condenser may be disposed in an outdoor space to exchange heat with outside air and the evaporator may be disposed in an indoor space to exchange heat with indoor air.

(20) As shown in FIG. 1, a building has a wall **1** and a window **2**, and the wall **1** may include a hole, that is, an installation space. The air conditioner **100** according to an embodiment may be placed in the hole, that is, in the installation space of the wall **1**. In contrast to FIG. 1, the air conditioner **100** according to an embodiment may be placed in an installation space provided in an area where the window **2** is located.

(21) FIG. 2 is a schematic perspective view of air conditioner **100** inserted into a case **200**. The air conditioner **100** may be inserted into the case **200** through a front surface of the case **200** and may be fixed into the case **200**. Hereinafter, a forward direction or a front surface refers to a direction toward the indoor space or a surface in that direction with respect to the wall **1** (see FIG. 1) or the window **2** (see FIG. 1) and a rearward direction or a rear surface refers to a direction toward a side opposite to the indoor space, that is, an outdoor space or a surface in that direction with respect to the wall **1** (see FIG. 1) or the window **2** (see FIG. 1).

(22) The case **200** may have a hexahedral shape and a front portion defining an opening and may

include an inner space. The case **200** may define at least one opening to suction and/or discharge air on a rear surface thereof. In addition, the case **200** may be fixed to the installation space of the air conditioner. The case **200** may be disposed in the installation space of the air conditioner in advance or may be provided to the user with the air conditioner **100**.

(23) The air conditioner **100** may include a front panel **110**. The front panel **110** may include an indoor suction inlet **112** through which indoor air may be suctioned on a front surface thereof and an indoor discharge outlet **114** through which conditioned air may be discharged at an upper portion thereof. The air suctioned through the indoor suction inlet **112** may be heat-exchanged while passing through a main body of the air conditioner **100** and then discharged through the indoor discharge outlet **114**.

(24) The front panel **110** may include an interface opening **116** at an upper portion thereof. A display of a control box may be exposed through the interface opening **116**, may indicate operation information of the air conditioner, and may also receive an input of a user operation command and/or a user-desired temperature.

(25) FIG. **3** is a schematic perspective view of air conditioner **100**, viewed from the front. FIG. **4** is a perspective view of air conditioner **100**, viewed from the rear.

(26) The air conditioner **100** according to an embodiment may include a front panel **110** and a main body **120**. The main body **120** may include a base **122**, an indoor main body disposed on the base **122**, and an outdoor main body disposed on the base **122**. The front panel **110** of the air conditioner **100** in FIGS. **3** and **4** may be the same as described with respect to FIG. **2**.

(27) The main body **120** may include base **122** and a plurality of components disposed on the base **122**. The plurality of components may include the indoor main body and the outdoor main body. The plurality of components may include a barrier assembly **136**, a control box **140**, a shroud **166**, a compressor **172**, and a condenser **190**. The main body **120** may be detachably coupled to the case **200**. Accordingly, the main body **120** may be separated from the case **200** for replacement or repair of components.

(28) As shown in FIG. **4**, the control box **140** may define an air suction inlet **141a** at a side thereof and an air discharge outlet **141b** on a rear surface thereof. The control box **140** may be disposed at a boundary line between the indoor main body and the outdoor main body and define the air suction inlet **141a** and the air discharge outlet **141b** to prevent a possibility of penetration of rainwater into the control box **140** and circulate outdoor air in the control box **140**.

(29) Detailed functions and arrangements of components of the air conditioner **100** shown in FIGS. **3** and **4** are described hereinafter with reference to FIG. **5**.

(30) FIG. **5** is a schematic exploded perspective view of air conditioner **100**. The air conditioner **100** according to an embodiment may include front panel **110** and main body **120**. The main body **120** may include base **122**, an evaporator **132**, a heater **134**, barrier assembly **136**, control box **140**, a control box cover **158**, a motor bracket **162**, a condenser fan motor **164**, shroud **166**, a condenser fan **168**, compressor **172**, a plurality of pipes **182**, **184**, and **186**, condenser **190**, and rear grills **192** and **194**.

(31) The front panel **110** may be the same as described with respect to FIG. **2**. The main body **120** may include base **122**, the indoor main body, and the outdoor main body. The barrier assembly **136** and the control box **140** may both be disposed at a boundary between the outdoor main body and the indoor main body. A rear surface of the barrier assembly **136** and a main substrate inside of the control box **140** may form a boundary between the outdoor main body and the main indoor body, components disposed in front of the rear surface of the barrier assembly **136** and the main substrate inside of the control box **140** may constitute the indoor main body, and components disposed behind the rear surface of the barrier assembly **136** and the main substrate inside of the control box **140** may constitute the outdoor main body.

(32) The evaporator **132** may be disposed on an inner surface of the front panel **110**. Air suctioned through the indoor suction inlet **112** may pass through the evaporator **132** and be discharged

through the indoor discharge outlet **114**. Refrigerant condensed by the condenser **190** may be introduced into the evaporator **132** through a third pipe **186**. The refrigerant may pass through the evaporator **132** and may be discharged to the compressor **172** through a first pipe **182**. The third pipe **186** may include an expansion device configured to decompress the refrigerant condensed by the condenser **190**. The air may be suctioned through the indoor suction inlet **112** and may be heat-exchanged with the refrigerant in the evaporator **132** while passing through the evaporator **132**. A heater **134** may be disposed on a rear surface of the evaporator **132** to heat the air suctioned through the indoor suction inlet **112**.

(33) The barrier assembly **136** may be disposed behind the evaporator **132**, that is, on the rear surface of the heater **134** disposed on the rear surface of the evaporator **132**, to partition the outdoor main body and the indoor main body. That is, a rear surface **136a** of the barrier assembly **136** may have a flat plate shape such that an indoor air flow path may be separated from an outdoor air flow path. The barrier assembly **136** may include an evaporator fan **136b** and an evaporator fan motor **136c**. The evaporator fan **136b** and the evaporator fan motor **136c** may each be disposed in front of the rear surface **136a** of the barrier assembly **136**. The evaporator fan motor **136c** may rotate the evaporator fan **136b**. The evaporator fan **136b** may discharge the air forward, which was introduced from the front. For example, the evaporator fan **136b** may suction the air that has passed through the evaporator **132** and/or the heater **134** disposed in front of the evaporator fan **136b** and discharge the suctioned air in a forward direction. The air discharged by the evaporator fan **136b** may be discharged to the indoor space through the indoor discharge outlet **114**.

(34) At least a portion of a front surface of the control box **140** may be disposed on the inner surface of the front panel **110** and the control box **140** may also be disposed at a side surface of the barrier assembly **136**. In addition, the control box **140** may be spaced apart from an inner surface of the case **200** (see FIG. 2) by a predetermined distance or more. For example, a distance between a side surface of the case **200** (see FIG. 2) and an inner surface of the control box **140** may be equal to or greater than 15 mm. The control box **140** may control the evaporator fan motor **136c**, the condenser fan motor **164**, and the compressor **172**. A configuration of the control box **140** is illustrated in FIGS. 6 to 9.

(35) Control box cover **158** may be disposed on the control box **140** to cover an upper surface of the control box **140**.

(36) The motor bracket **162** may be disposed on a front surface of the condenser fan motor **164** to support and fix the condenser fan motor **164**.

(37) The condenser fan motor **164** may be disposed between the rear surface of the barrier assembly **136** and the shroud **166**. For example, the condenser fan motor **164** may be arranged such that a center of a motor shaft of the condenser fan motor **164** corresponds to a center of a shroud opening **166a**.

(38) The shroud **166** may be disposed between the barrier assembly **136** and the condenser **190**. The shroud **166** may be disposed behind the barrier assembly **136** and may be spaced apart from the barrier assembly **136**. In addition, the shroud **166** may be coupled to the condenser **190**. The shroud **166** may also be coupled to the condenser fan **168** to guide a flow of air passing through the condenser fan **168**. The shroud **166** may include the shroud opening **166a** to guide air to a suction side of the condenser fan **168**. That is, outdoor air introduced into the case **200** (see FIG. 2) may be suctioned into the condenser fan **168** through the shroud opening **166a**.

(39) The condenser fan **168** may be disposed between the shroud **166** and the condenser **190**. The condenser fan **168** may be disposed in a space formed by the shroud **166** in front of the condenser **190**. The condenser fan **168** may be coupled to the shroud **166** to suction air through the shroud opening **166a** and discharge the suctioned air to the condenser **190**. That is, the condenser fan **168** may blow outdoor air. The condenser fan **168** may be an axial-flow fan to suction air in an axial direction and discharge the air in the axial direction.

(40) The compressor **172** may be disposed in a space between a rear of the control box **140** and a

side of the shroud **166**. The compressor **172** may be spaced apart from each of the control box **140** and the shroud **166**. A gas-liquid separator may be disposed at one side of the compressor **172**. The compressor **172** may be connected to the evaporator **132** through the first pipe **182**. The refrigerant discharged from the evaporator **132** may flow into the compressor **172** through the first pipe **182**. The compressor **172** may compress the introduced refrigerant and discharge the refrigerant to the condenser **190** through the second pipe **184**.

(41) The condenser **190** may be disposed on a rear surface of the shroud **166**. The condenser **190** may be coupled to the shroud **166** on the rear surface of the shroud **166**. Air discharged from the condenser fan **168** may be discharged to the outdoor space through the condenser **190**. Refrigerant discharged from the compressor **172** may flow into the condenser **190** through the second pipe **184**. The condenser **190** may condense the introduced refrigerant and discharge the refrigerant to the evaporator **132** through the third pipe **186**. Rear grills **192** and **194** may be disposed on a rear surface or both sides of the condenser **190** and guide air that has passed through the condenser **190**.

(42) FIG. **6** is a schematic perspective view of control box **140** of air conditioner **100** according to an embodiment. The control box **140** may include a barrier **141**, a reactor cover **148**, a first power cover **149**, and a display cover **151**.

(43) At least a portion of the control box **140** may be disposed on an inner side surface of the front panel **110** (see FIGS. **2** and **3**). For example, the reactor cover **148**, the first power cover **149**, and the display cover **151** of the control box **140** may be disposed on the inner side surface of the front panel **110** (see FIGS. **2** and **3**). The barrier **141** of the control box **140** may be disposed behind a rear end of the front panel **110** (see FIGS. **2** and **3**). In addition, the barrier **141** of the control box **140** may define air suction inlet **141a** to suction air at a side surface thereof.

(44) FIG. **7** is a schematic exploded perspective view of control box **140** of air conditioner **100** in FIG. **6**. The control box **140** may include barrier **141**, a control box case **142**, a main substrate **143**, a heat sink **144**, a control box cover **145**, a reactor base **146**, a reactor **147**, reactor cover **148**, first power cover **149**, a second power cover **150**, a display including display cover **151**, a display substrate **152**, and a communication module **153**, a sub cover **154**, a sub-substrate **155**, a supporter **156**, and a transformer **157**.

(45) The barrier **141** may define a first side surface and a rear surface of an appearance of the control box **140** and may support and protect other components of the control box **140**. The barrier **141** may be made of metal, for example. The barrier **141** may be disposed at a rear surface of the control box case **142** and be coupled to the control box cover **145** to cover the rear surface and the first side surface of the control box case **142**. The barrier **141** may define air suction inlet **141a** through which air is suctioned into the control box **140** and air discharge outlet **141b** through which air inside of the control box **140** is discharged. The air discharge outlet **141b** may have a size larger than a size of the air suction inlet **141a** such that the air smoothly flows and may be defined on a first surface of the barrier **141** and the air suction inlet **141a** may be defined on a second surface that is bent and extends from the first surface thereof. The first surface may form a rear surface of the control box **140** and the second surface may form a first side surface of the control box **140**. The first side surface of the control box **140** may face the inner surface of the case **200** (see FIG. **2**). In addition, the air suction inlet **141a** and the air discharge outlet **141b** of the barrier **141** may be provided at a portion surrounding an area of the heat sink.

(46) The control box case **142** may be coupled to the barrier **141**. The control box case **142** may be made of plastic and be formed by injection, for example. Alternatively, the control box case **142** may be made of metal, for example. The control box case **142** may be coupled to the control box cover **145** to support the main substrate **143** and include a partition **142a** to divide a rear space of the control box case **142** into a first area and a second area. The first area may be disposed under the second area. A heat sink mounted on the main substrate **143** may be disposed in the first area and elements mounted on the main substrate **143** may be disposed in the second area.

(47) The main substrate **143** may be coupled to the control box cover **145** and support various

elements to control the evaporator fan motor **136c** (see FIG. 5), the condenser fan motor **164** (see FIG. 5), and the compressor **172** (see FIG. 5), and the heat sink **144**, and may include a first substrate surface facing the front surface of the air conditioner **100** and a second substrate surface opposite to the first substrate surface to support the elements and the heat sink **144**.

(48) The heat sink **144** may be mounted on the main substrate **143** to dissipate heat generated from the elements mounted on the main substrate **143**. The heat sink **144** may include a plurality of plates disposed parallel to one another. Each of the plurality of plates may be disposed to be perpendicular to the substrate. The air suction inlet **141a** may be disposed at a side of the heat sink **144** and the air discharge outlet **141b** may be disposed at a rear side of the heat sink **144**.

(49) The control box cover **145** may cover a surface of the main substrate **143**. For example, the control box cover **145** may cover the second substrate surface of the main substrate **142**. The main substrate **143** may be coupled to the rear surface of the control box cover **145**.

(50) The reactor base **146** may be coupled to the control box cover **145**, may be disposed at a front surface of the control box cover **145**, and may fix the reactor. The reactor **147** may be coupled to the reactor base **146** and may be disposed on a front surface of the reactor base **146**. The reactor cover **148** may be coupled to the reactor base **146** to cover the reactor **147** and may have a hexahedral shape and define an opening at a rear surface thereof.

(51) The second power cover **150** may be coupled to a front surface of the control box cover **145** and the first power cover **149** may be coupled to the second power cover **150**. The first power cover **149** and the second power cover **150** may cover a power-supply terminal.

(52) The display cover **151** may be disposed at an upper portion of a front surface of the control box **140** and may be exposed to outside through the interface opening **116** of the front panel **110** (see FIGS. 2 and 3). The display substrate **152** may be coupled to an inner surface of the display cover **151** to support elements for indicating a state of the air conditioner and elements for reviewing input of a user's command or setting information. The user command may include, for example, an operation start command and an operation end command of the air conditioner and the setting information may include, for example, desired temperature information and/or intensity information of discharged air.

(53) The communication module **153** may be mounted on the display substrate **152** or may be coupled to the display cover **151**. The communication module **153** may transmit and receive data to and from an external communication device and may be, for example, a wireless fidelity (Wi-Fi) device.

(54) The sub cover **154** may be coupled to the front surface of the control box cover **145** to support and fix the sub-substrate **155**. The sub-substrate **155** may be coupled to the sub cover **154** to place elements for supporting functions performed by the elements mounted on the main substrate **143** or performing additional functions in addition to the functions performed by the elements mounted on the main substrate **143**. The supporter **156** may be coupled to the front surface of the control box cover **145** to prevent the display cover **151** from sagging downward.

(55) The transformer **157** may be mounted on the main substrate **143** or may be mounted at an additional location, such as the control box cover **145**. The transformer **157** may convert an external-input power into a required power.

(56) FIG. 8 is a schematic side view of control box **140** of air conditioner **100** according to an embodiment. In FIG. 8, alternated long and short dash line "L1" refers to a position where the main substrate **143** (see FIG. 7) is provided and main body **120** of air conditioner **100** may be divided into an indoor main body and an outdoor main body with respect to the alternated long and short dash line "L1".

(57) That is, according to embodiments disclosed herein, the main body **120** of the air conditioner **100** may include the indoor main body to blow indoor air and the outdoor main body to blow outdoor air. In addition, according to embodiments disclosed herein, the control box **140** may be disposed at a boundary between the outdoor main body and the indoor main body. For example, the

main body may be divided into the outdoor main body and the indoor main body with respect to the main substrate **143** (see FIG. 7) of the control box **140**. That is, the outdoor air may flow in a space behind the main substrate **143** (see FIG. 7) of the control box **140**.

(58) FIGS. **9** to **11** show examples of air flow at an outdoor portion of an air conditioner according to an embodiment. In FIGS. **9** to **11**, arrows indicate flow of outdoor air. FIG. **9** is a schematic perspective view of control box **140** of air conditioner **100** viewed from the rear and the side.

(59) The control box **140** may include air suction inlet **141a** to introduce air into the control box **140** at a first side surface thereof and air discharge outlet **141b** to discharge air from the control box **140** on a rear surface thereof. In this case, the first side surface of the control box **140** faces the case **200** (see FIG. 2).

(60) FIG. **10** is a cross-sectional view showing an air conditioner taken along line X-X' in FIG. 2. FIG. **11** is a perspective view of air conditioner being inserted into a case, when viewed from the rear and the side.

(61) As shown in FIGS. **4** and **5**, the control box **140** of the air conditioner **100** according to embodiments disclosed herein may be disposed on a side surface of the barrier assembly **136**. As shown, the rear surface of the control box **140**, that is, the barrier **141** (see FIGS. **6** to **9**) of the control box **140**) may be flush with the rear surface of the barrier assembly **136**. Alternatively, in contrast to the drawings, the rear surface of the control box **140** and the rear surface of the barrier assembly **136** may be disposed on different planes.

(62) The main body may be divided into the indoor main body and the outdoor main body with respect to the rear surface **136a** of the barrier assembly **136**. The indoor air may flow based on an operation of the evaporator fan **136b** in a front portion of the main body with respect to the rear surface **136a** of the barrier assembly **136**. For example, the indoor air may be suctioned through the indoor suction inlet **112** of the front panel **110**, pass through the evaporator **132** (see FIG. 5) and/or the heater **134** (see FIG. 5), and be suctioned into the evaporator fan **136b**. Subsequently, the air may be discharged from the evaporator fan **136b** and then discharged through the indoor discharge outlet **114** of the front panel **110**.

(63) In addition, outdoor air may flow based on an operation of the condenser fan **168** in a rear space of the main body with respect to the rear surface **136a** of the barrier assembly **136**. For example, the outdoor air may be introduced into the case **200** through a space between the condenser **190** and the case **200**. Subsequently, the introduced air may pass through the main body, that is, the outdoor main body, of the air conditioner inside of the case **200**, and through the shroud opening **166a** (see FIG. 5) of the shroud **166**, and be suctioned to the condenser fan **168** (see FIG. 5). Subsequently, the air may be discharged from the condenser fan **168** (see FIG. 5) and be discharged to outside through the condenser **190**.

(64) As the control box **140** is disposed on the side surface of the barrier assembly **136**, and the shroud **166** and the condenser fan **168** are disposed behind the barrier assembly **136**, an air flow path may be defined behind the control box **140**. For example, outdoor air introduced into the space between the case **200** and the condenser **190** may flow into the shroud opening **166a** through a space between the control box **140** and the compressor **172**.

(65) As the air flow path is defined behind the control box **140**, pressure of the air discharge outlet **141b** of the control box **140** may be lower than the pressure inside of the control box **140**, and thus, the air inside of the control box **140** may be discharged through the air discharge outlet **141b**. Further, as the air inside of the control box **140** is discharged, air may be suctioned into the air suction inlet **141a** defined at the first side surface of the control box **140**. For example, the outdoor air introduced into the space between the case **200** and the condenser **190** may pass through the space between the control box **140** and the case **200** and may be introduced into the air suction inlet **141a**.

(66) According to embodiments disclosed herein, there is provided an air conditioner capable of preventing direct exposure of a control box and/or a heat sink of the control box to rainwater even

if it rains. According to embodiments disclosed herein, there is further provided an air conditioner capable of facilitating heat dissipation of control components disposed inside a control box to control an inverter.

(67) The objects are not limited to the above-mentioned objects, and other objects and advantages which are not mentioned may be understood by the description and more clearly understood based on embodiments. It will also be readily understood that the objects and the advantages may be implemented by features defined in claims and a combination thereof.

(68) According to embodiments disclosed herein, the air conditioner may include a control box. The control box may be disposed at a boundary between an indoor main body to blow indoor air and an outdoor main body to blow outdoor air and may introduce the outdoor air to an inside thereof.

(69) The air conditioner may include a barrier assembly including an evaporator fan, an evaporator disposed in front of the barrier assembly, a shroud disposed behind the barrier assembly and defining a shroud opening, a condenser, and a condenser fan disposed between the shroud opening and the condenser. The control box may be disposed on or at a side surface of the barrier assembly.

(70) The control box may have a first side surface spaced apart from the case, defining an air suction inlet configured to suction air on the first side surface thereof and an air discharge outlet configured to discharge the air on a rear surface thereof. Further, the control box may include a main substrate and a heat sink mounted on the main substrate. The air suction inlet and the air discharge outlet may be defined at a portion covering a position area of the heat sink.

(71) The air conditioner may include a base; a barrier assembly disposed on the base and including an evaporator fan configured to introduce air from a front and discharge the introduced air; an evaporator disposed in front of the barrier assembly; a shroud disposed behind the barrier assembly, spaced apart from the barrier assembly, and defining a shroud opening; a condenser disposed behind the shroud; a condenser fan disposed between the shroud opening and the condenser; a compressor connected to the evaporator and the condenser through a pipe; and a control box configured to control the compressor. The control box may be disposed on a side surface of the barrier assembly, have a first side surface spaced apart from the case, define an air suction inlet configured to suction air into the control box on the first side surface thereof and an air discharge outlet configured to discharge the air inside the control box on a rear surface thereof.

(72) The control box may include a main substrate configured to support elements to control the compressor and a heat sink mounted on the main substrate. The heat sink may be disposed in a space between the air suction inlet and the air discharge outlet.

(73) The heat sink may include a plurality of flat plates. Each of the plurality of flat plates may be disposed to be perpendicular to the main substrate.

(74) The control box may include a control box cover disposed on a front surface of the main substrate and coupled to the main substrate; a control box case disposed on a rear surface of the control box cover and coupled to the control box cover; and a barrier disposed on a rear surface of the control box case, coupled to the control box cover, defining the first side surface and the rear surface of an appearance of the control box, and the air suction inlet and the air discharge outlet.

(75) The control box case may include a partition configured to divide a rear space of the control box case into a first area and a second area. The heat sink may be mounted on the main substrate and be disposed in the second area, and the air suction inlet and the air discharge outlet may be each defined in the barrier and surround the second area.

(76) The barrier may include a first surface defining a rear appearance of the control box and the air discharge outlet, and a second surface that is bent and extends from the first surface, defining an appearance of a first side surface of the control box and the air suction inlet.

(77) The control box may further include a reactor base disposed on a front surface of the control box cover and coupled to the control box cover; a reactor disposed on a front surface of the reactor base and coupled to the reactor base; and a reactor cover disposed on a front surface of the control

box cover, coupled to the control box cover, and configured to cover the reactor.

(78) The air conditioner may further include a front panel defining a front appearance of the air conditioner, an indoor suction inlet configured to suction indoor air and an indoor discharge outlet configured to discharge the air, which was discharged from the evaporator fan to an indoor space, and in which at least a portion of the control box and the evaporator are each disposed inside of the front panel.

(79) The control box may include a display configured to receive command and setting information from a user and indicate a state of the air conditioner. The display may be exposed to outside through an interface opening defined at the front panel.

(80) The air discharge outlet may have a size larger than a size of the air suction inlet. Further, a distance between the case and the first side surface of the control box may be equal to or greater than 15 mm.

(81) The compressor may be disposed behind the control box. Further, the compressor may be disposed beside the shroud.

(82) A rear surface of the control box may be flush with a rear surface of the barrier assembly.

(83) According to embodiments disclosed herein, even if it rains, the air conditioner may prevent direct exposure of the control box and/or the heat sink inside the control box to rainwater. Further, according to an embodiment disclosed herein, the air conditioner may facilitate heat dissipation of control components disposed in the control box configured to control an inverter.

(84) Although embodiments have been described with reference to drawings hereinabove, the embodiments are not limited to such embodiments and the drawings herein, and various modifications can be made by the skilled person in the art within the scope of the technical idea. Further, even if working effects obtained based on configurations are not explicitly described in the description of embodiments, effects predictable based on the corresponding configuration have to be recognized.

(85) It will be understood that when an element or layer is referred to as being “on” another element or layer, the element or layer can be directly on another element or layer or intervening elements or layers. In contrast, when an element is referred to as being “directly on” another element or layer, there are no intervening elements or layers present. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

(86) It will be understood that, although the terms first, second, third, etc., may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer or section from another region, layer or section. Thus, a first element, component, region, layer or section could be termed a second element, component, region, layer or section without departing from the teachings of the present invention.

(87) Spatially relative terms, such as “lower”, “upper” and the like, may be used herein for ease of description to describe the relationship of one element or feature to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation, in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “lower” relative to other elements or features would then be oriented “upper” relative to the other elements or features. Thus, the exemplary term “lower” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

(88) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used

in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(89) Embodiments of the disclosure are described herein with reference to cross-section illustrations that are schematic illustrations of idealized embodiments (and intermediate structures) of the disclosure. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments of the disclosure should not be construed as limited to the particular shapes of regions illustrated herein but are to include deviations in shapes that result, for example, from manufacturing.

(90) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(91) Any reference in this specification to “one embodiment,” “an embodiment,” “example embodiment,” etc., means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. The appearances of such phrases in various places in the specification are not necessarily all referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with any embodiment, it is submitted that it is within the purview of one skilled in the art to effect such feature, structure, or characteristic in connection with other ones of the embodiments.

(92) Although embodiments have been described with reference to a number of illustrative embodiments thereof, it should be understood that numerous other modifications and embodiments can be devised by those skilled in the art that will fall within the spirit and scope of the principles of this disclosure. More particularly, various variations and modifications are possible in the component parts and/or arrangements of the subject combination arrangement within the scope of the disclosure, the drawings and the appended claims. In addition to variations and modifications in the component parts and/or arrangements, alternative uses will also be apparent to those skilled in the art.

Claims

1. An air conditioner, comprising: a base; a case disposed on the base; a barrier assembly disposed on the base and comprising an evaporator fan configured to introduce air from a front and discharge the introduced air; an evaporator disposed in front of the barrier assembly; a shroud disposed behind the barrier assembly, spaced apart from the barrier assembly, and defining a shroud opening; a condenser disposed behind the shroud; a condenser fan disposed between the shroud opening and the condenser; a compressor connected to the evaporator and the condenser through a pipe; and a control box configured to control the compressor, wherein the control box is disposed at a left side surface or at a right side surface of the barrier assembly and comprises: a control box case; a control box cover coupled to the control box case; a main substrate disposed between the control box case and the control box cover; a heat sink mounted on the main substrate; and a barrier disposed at a rear surface of the control box case, wherein the barrier has a first side surface spaced apart from the case, and defines an air suction inlet through which air is suctioned into the control box on the first side surface thereof and an air discharge outlet through which the air inside the control box is discharged on a rear surface thereof, and wherein the barrier assembly and the control box are disposed at a boundary between an outdoor body and an indoor body.
2. The air conditioner of claim 1, wherein: the heat sink is disposed in a space between the air

suction inlet and the air discharge outlet.

3. The air conditioner of claim 2, wherein the heat sink comprises a plurality of flat plates, and each of the plurality of flat plates extends perpendicular to the main substrate.

4. The air conditioner of claim 1, wherein the barrier is coupled to the control box cover, and defines the first side surface and the rear surface of the control box, and the air suction inlet and the air discharge outlet.

5. The air conditioner of claim 1, wherein the control box case comprises a partition configured to divide a rear space of the control box case into a first area and a second area, where the heat sink is mounted on the main substrate and is disposed in the second area, and wherein the air suction inlet and the air discharge outlet are each defined in the barrier and surround the second area.

6. The air conditioner of claim 1, wherein the barrier comprises: a first surface defining the rear surface of the control box and the air discharge outlet; and a second surface bent and extended from the first surface, defining the first side surface of the control box and the air suction inlet.

7. The air conditioner of claim 1, wherein the control box further comprises: a reactor base disposed on a front surface of the control box cover and coupled to the control box cover; a reactor disposed on a front surface of the reactor base and coupled to the reactor base; and a reactor cover disposed on the front surface of the control box cover, coupled to the control box cover, and configured to cover the reactor.

8. The air conditioner of claim 1, further comprising a front panel defining a front appearance of the air conditioner, an indoor suction inlet through which indoor air is suctioned and an indoor discharge outlet through which the air discharged from the evaporator fan is discharged to an indoor space, and in which at least a portion of the control box and the evaporator are each disposed inside of the front panel.

9. The air conditioner of claim 8, wherein the control box comprises a display configured to receive command and setting information from a user and indicate a state of the air conditioner, and wherein the display is exposed to an outside through an interface opening defined at the front panel.

10. The air conditioner of claim 1, wherein the air discharge outlet has a size larger than a size of the air suction inlet.

11. The air conditioner of claim 1, wherein a distance between the case and the first side surface of the control box is equal to or greater than 15 mm.

12. The air conditioner of claim 1, wherein the compressor is disposed behind the control box.

13. The air conditioner of claim 12, wherein the compressor is disposed adjacent to the shroud.

14. The air conditioner of claim 1, wherein the rear surface of the control box is flush with a rear surface of the barrier assembly.

15. An air conditioner, comprising: a base; a case forming an outer appearance of the air conditioner, and disposed on the base; a barrier assembly disposed on the base and comprising an evaporator fan configured to introduce air into the air conditioner and discharge the introduced air; an evaporator disposed at a first side of the barrier assembly; a shroud disposed at a second side of the barrier assembly, spaced apart from the barrier assembly, and defining a shroud opening; a condenser disposed adjacent to the shroud; a condenser fan disposed between the shroud opening and the condenser; a compressor connected to the evaporator and the condenser through a pipe; and a control box configured to control the compressor, wherein the control box is disposed at a third side of the barrier assembly and comprises: a control box case; a control box cover coupled to the control box case; a main substrate disposed between the control box case and the control box cover; a heat sink mounted on the main substrate; and a barrier disposed at a rear surface of the control box case, the third side being different from the first side and the second side, wherein the barrier has a first side surface spaced apart from the case, defines an air suction inlet through which air is suctioned into the control box on the first side surface thereof and an air discharge outlet through which the air inside the control box is discharged on a rear surface thereof, and comprises a heat

sink disposed in a space between the air suction inlet and the air discharge outlet, wherein the third side of the barrier assembly extends in a vertical direction between the first and second sides of the barrier assembly, and wherein the barrier assembly and the control box are disposed at a boundary between an outdoor body and an indoor body.

16. The air conditioner of claim 15, wherein the air discharge outlet has a size larger than a size of the air suction inlet.

17. The air conditioner of claim 15, wherein a distance between the case and the first side surface of the control box is equal to or greater than 15 mm.

18. The air conditioner of claim 15, wherein the compressor is disposed behind the control box.

19. The air conditioner of claim 18, wherein the compressor is disposed adjacent to the shroud.

20. The air conditioner of claim 15, wherein the rear surface of the control box is flush with a rear surface of the barrier assembly.
